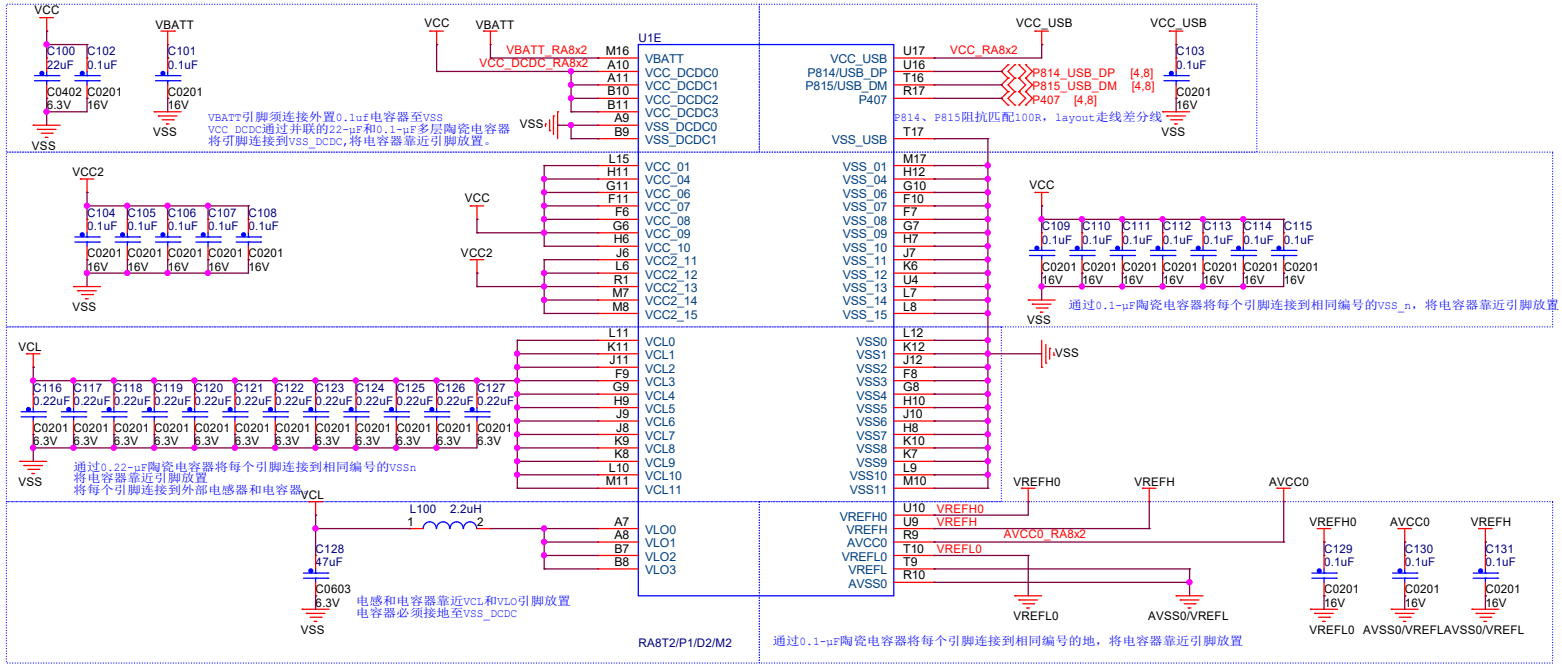
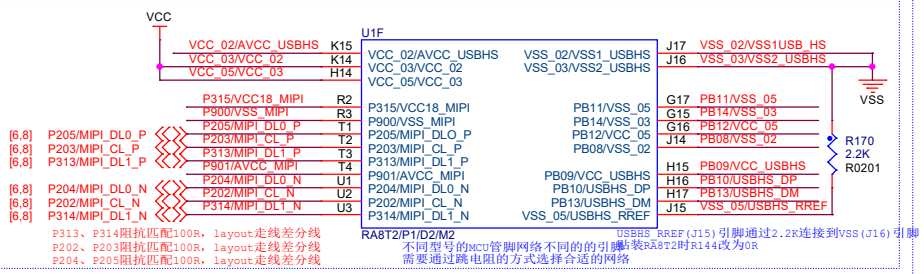
Page1: MCU 电源
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Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB



核心板缺省功能阻容设置

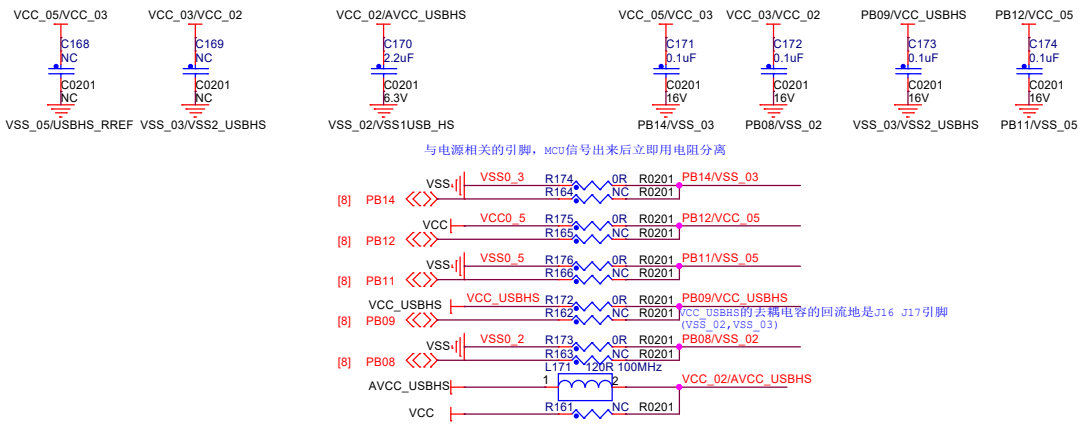
核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	0.1uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1UF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R@100M
R172-R180	NC	0R
R191-R193	NC	0R



由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2

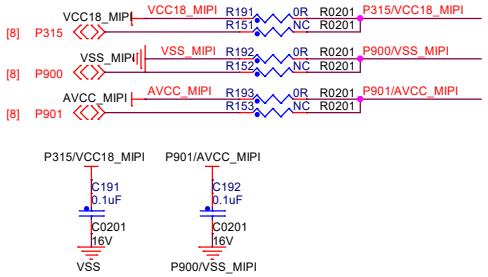
电阻选装

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放

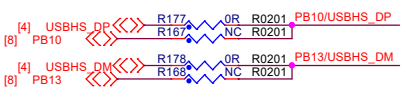


MIPI部分

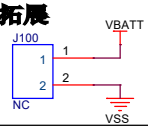
要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放



差分线走到J1(BTB)附近再用电阻分离。

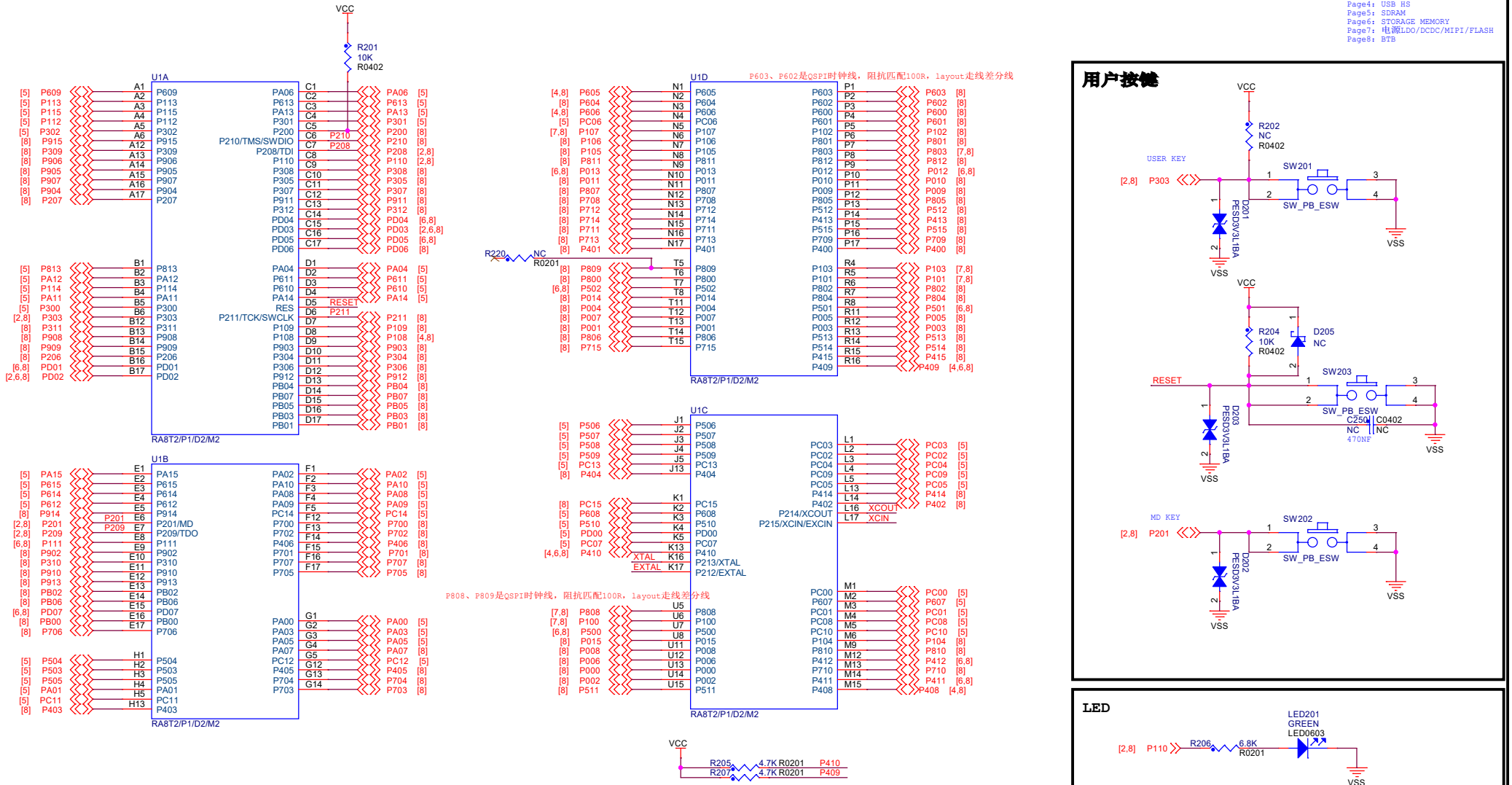


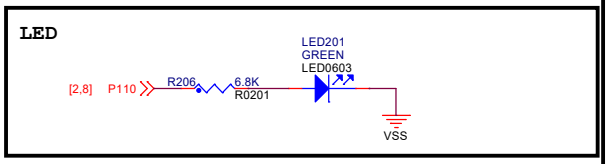
通孔拓展



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Size A3	Document Number MCU 电源		Rev V1.2
Date: Monday, June 08, 2026		Sheet 1 of 10	

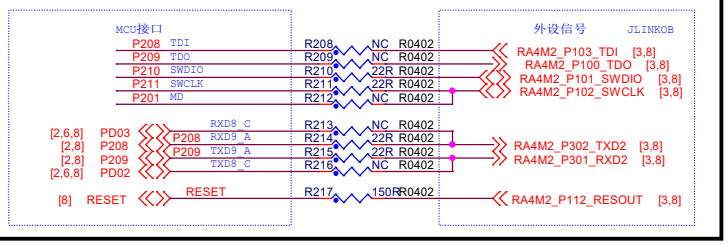
Page1: MCU 电源
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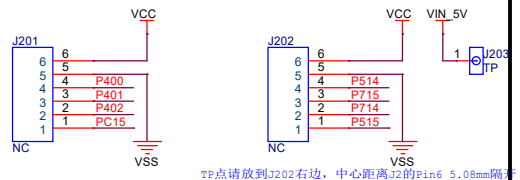
如果使用USB HS，主晶振只能选12/20/24/48MHz

接口信号



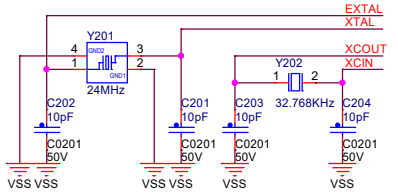
通孔拓展

VIN有大电流需求，请用粗线或铜皮连到VIN网络。
J201，J202之间的间距请保证2.54mm的倍数



TP点请放到J202右边，中心距离J2的Pin6 5.08mm隔开一个2.54mm管脚的位置（防止误接）

晶振



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Jlink OB

注意：R7FA4M2只能12MHz晶振

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MCU调试/下载接口

[2,8] RA4M2_P302_TXD2
[2,8] RA4M2_P301_RXD2
[2,8] RA4M2_P100_TDO
[2,8] RA4M2_P101_SWDIO
[2,8] RA4M2_P102_SWCLK
[2,8] RA4M2_P103_TDI
JLOB_RESOUT

TDO SWDIO SWCLK TDI

VCC_RA4M2

R306 30K R0201

U5

P000 P001 P002 P013 P014 P015 P100 P101 P102 P103 P104 P108/TMS/SWDIO P109/TDO/SWO P110/TDI P111 P112 P200 P201/MD P206 P207 P212/EXTAL P213/XTAL XCIN XCOUT VCL

P300/TCK/SWCLK P301 P302 P402 P407 P408 P409 P500 RES USB_DP USB_DM VBATT VCC1 VCC2 VCC_USB VREFH0 VREFH AVCC0 VSS1 VSS2 AVSS0/VREFL VREFL0 VSS_USB EPAD

R7FA4M2AD3CNE

LED301 YELLOW LED0603

DBG_USBFS_VBUS
[2,8] RA4M2_P112_RESOUT
R311 10K R0201

VCC_RA4M2 VCC_RA4M2
R313 10K R0201
R314 10K R0201
4M2_MD

Y301 12MHz
C309 18pF C0201 50V
C310 18pF C0201 50V
R317 10K R0201
C311 0.22uF C0201 6.3V

VSS_RA4M2

MD	MODE
1	Single-chip mode
0	SCI/USB boot mode

4M2_RES
R301 10K R0201
R305 NC R0402
4M2_P407
R319 0R R0402
R320 0R R0402
DBG_USB_DP
DBG_USB_DM
DBG_USB_VBUS [4]

VCC_5V
R303 15K R0201
R307 30K R0201
D301 B340A
DBG_USBFS_VBUS

JDBG TYPE-C
GND1 VBUS1 CC1 DP1 DN1 SBU1 VBUS2 GND2 GND3 VBUS3 SBU2 DN2 DP2 CC2 VBUS4 GND4 SH1 SH2 SH3 SH4
C1 C2 C3 C4
JDBG_EGND

R315 5.1K R0201
R316 5.1K R0201

U302 SR05-N
DM DP VBUS GND

layout：USB差分线控阻抗90R：
DBG_USB_DP/DBG_USB_DM

测试点 1.25间距/预留

J301 NC
DBG_USBFS_VBUS
VCC_5V

[8] RA4M2_RESET
4M2_MD
4M2_RES
J302 NC

debug隔离工作时，用作短路二极管运行

VCC_RA4M2
C306 10uF C0402 10V
C301 0.1uF C0201 16V
C302 0.1uF C0201 16V
C307 0.1uF C0201 16V
C303 0.1uF C0201 16V
C308 0.1uF C0201 16V
C304 0.1uF C0201 16V
C305 0.1uF C0201 16V

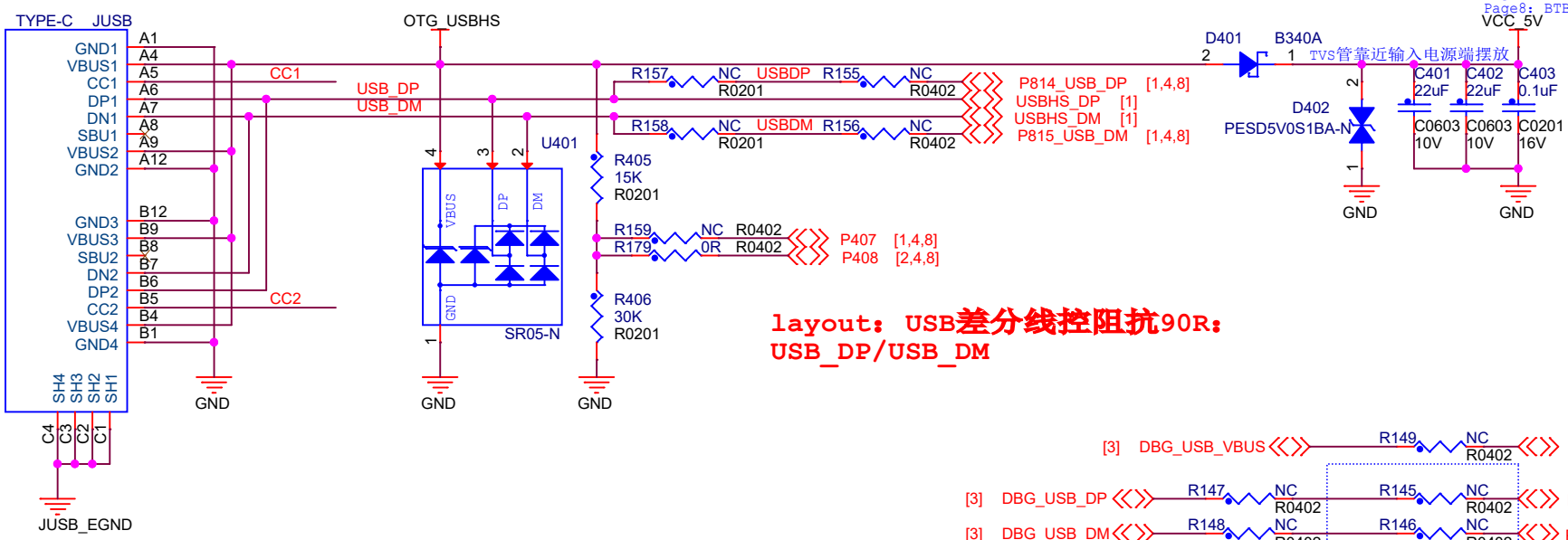
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Size A4	Document Number Jlink OB	Rev V1.2
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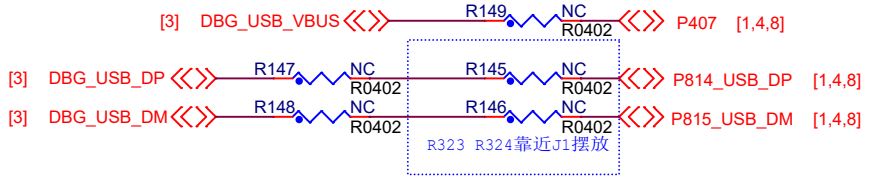
USB HS

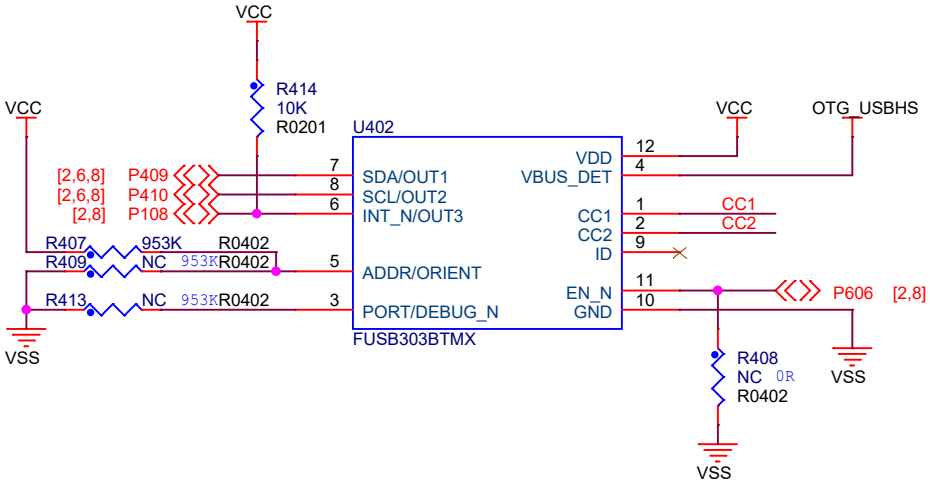
如果使用USB HS，主晶振只能选12/20/24/48MHz

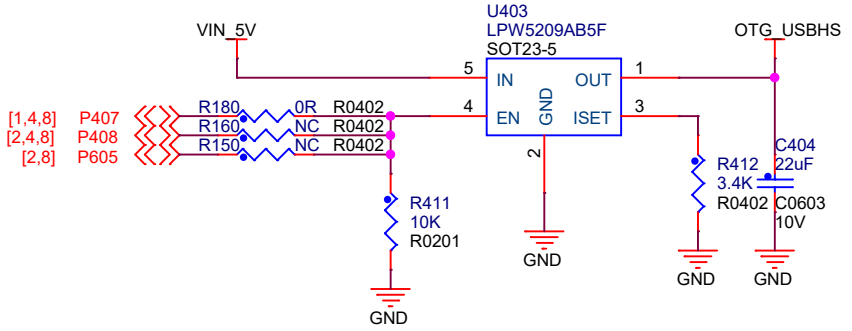
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layout：USB差分线控阻抗90R：
USB_DP/USB_DM





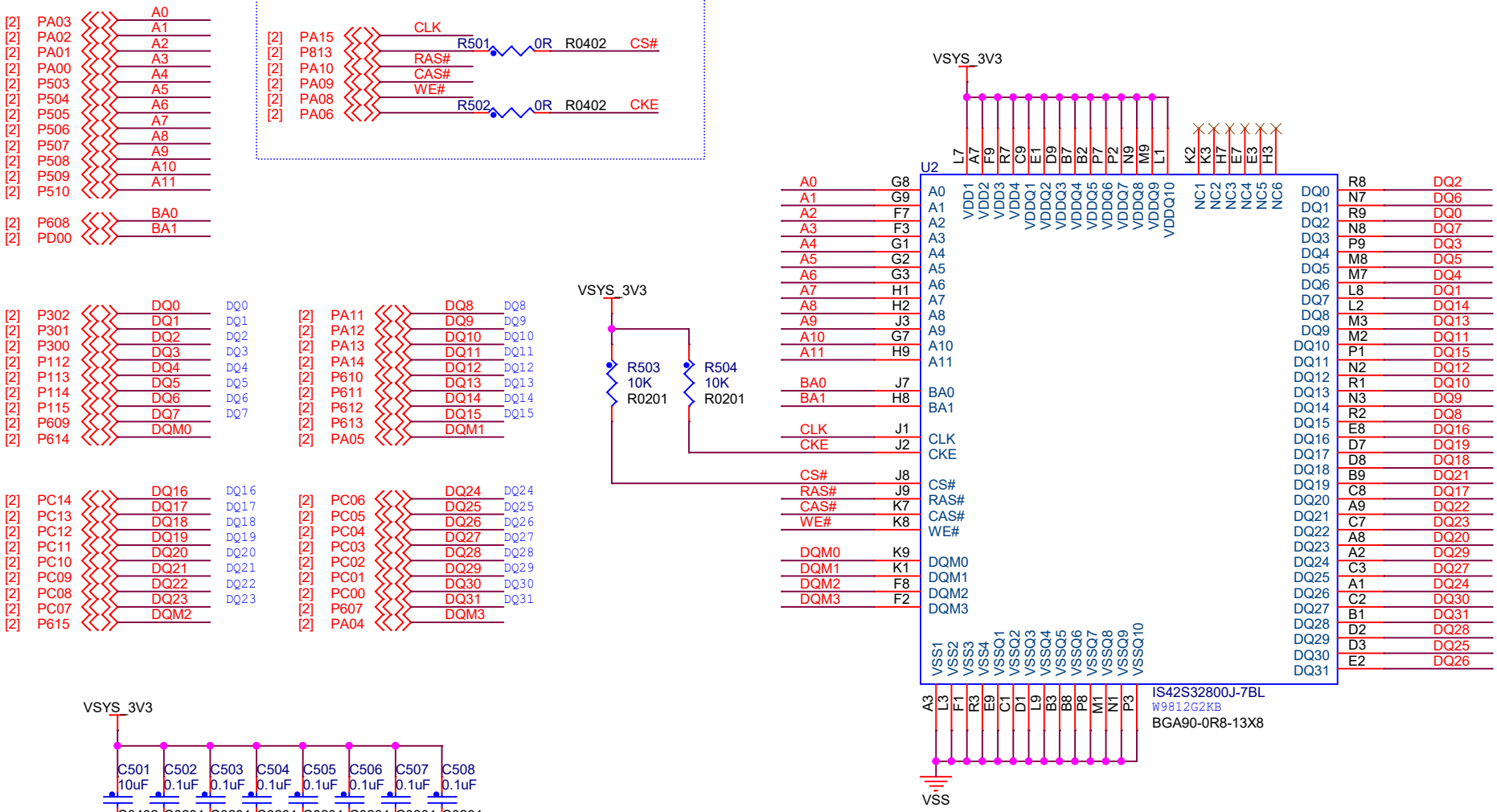


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SDRAM

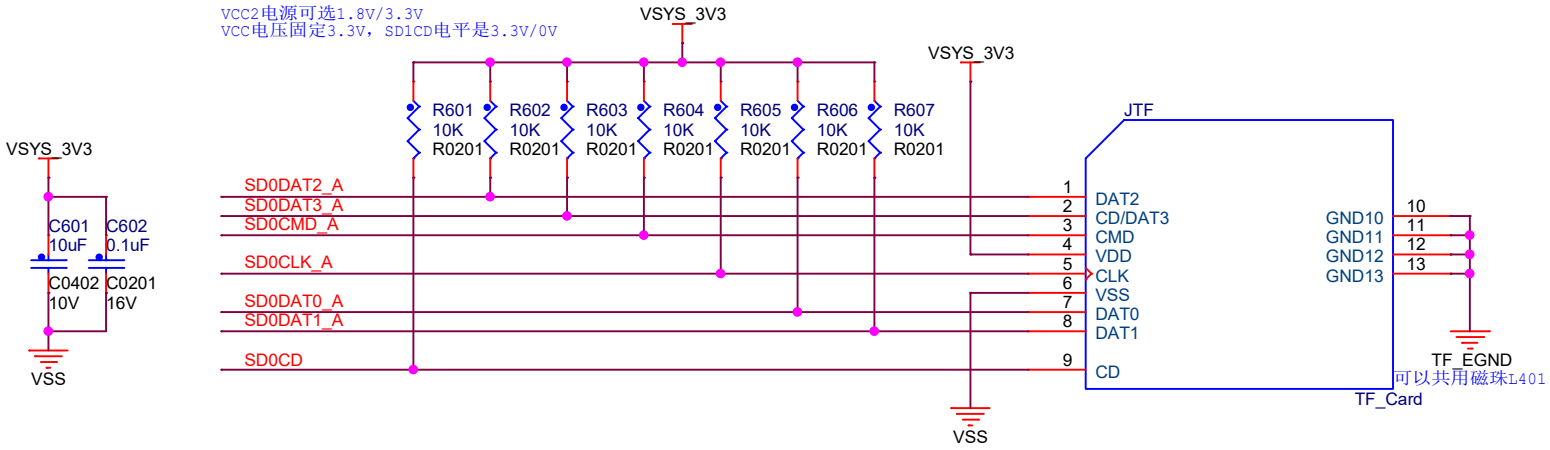
Page1: MCU 电源
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layout： SDRAM走线控阻抗50R：



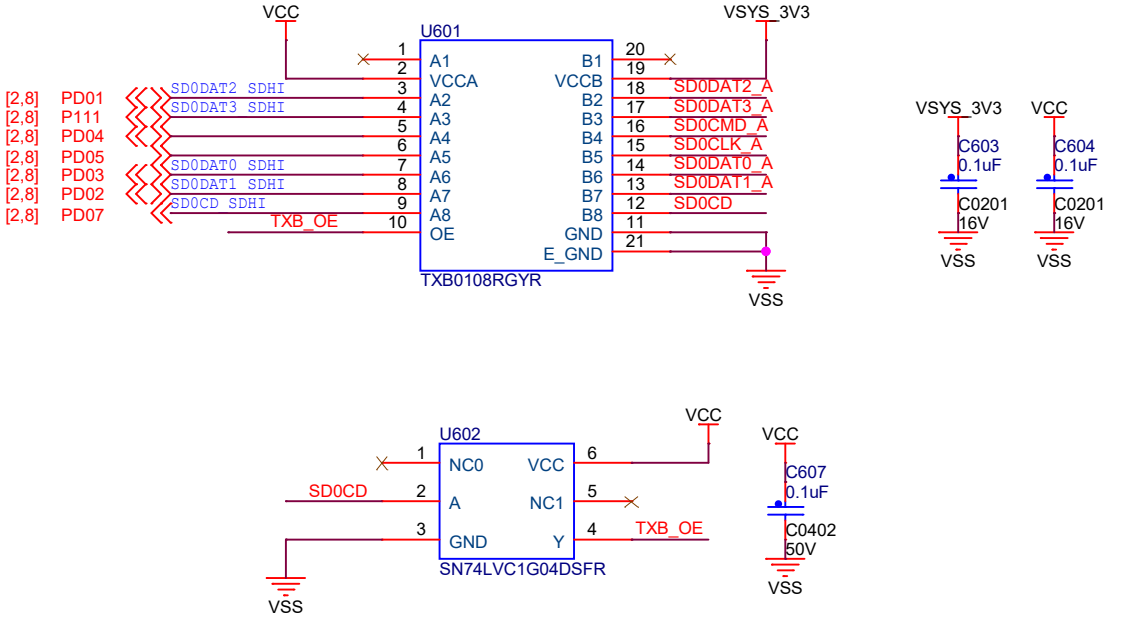
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Size: A4	Document Number: SDRAM	Rev: V1.2
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SD卡

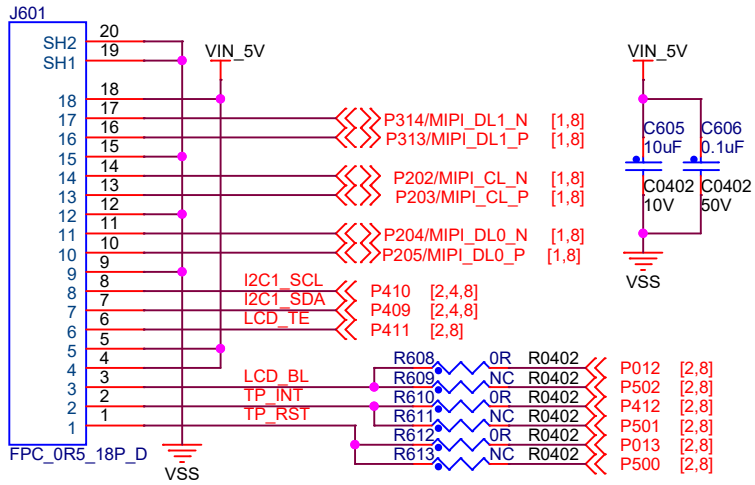


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SD卡电平转换

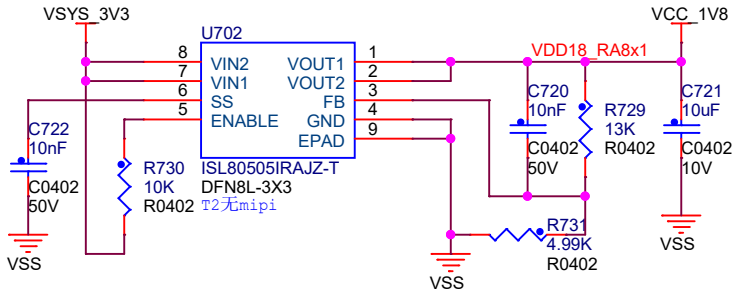


MIPI屏

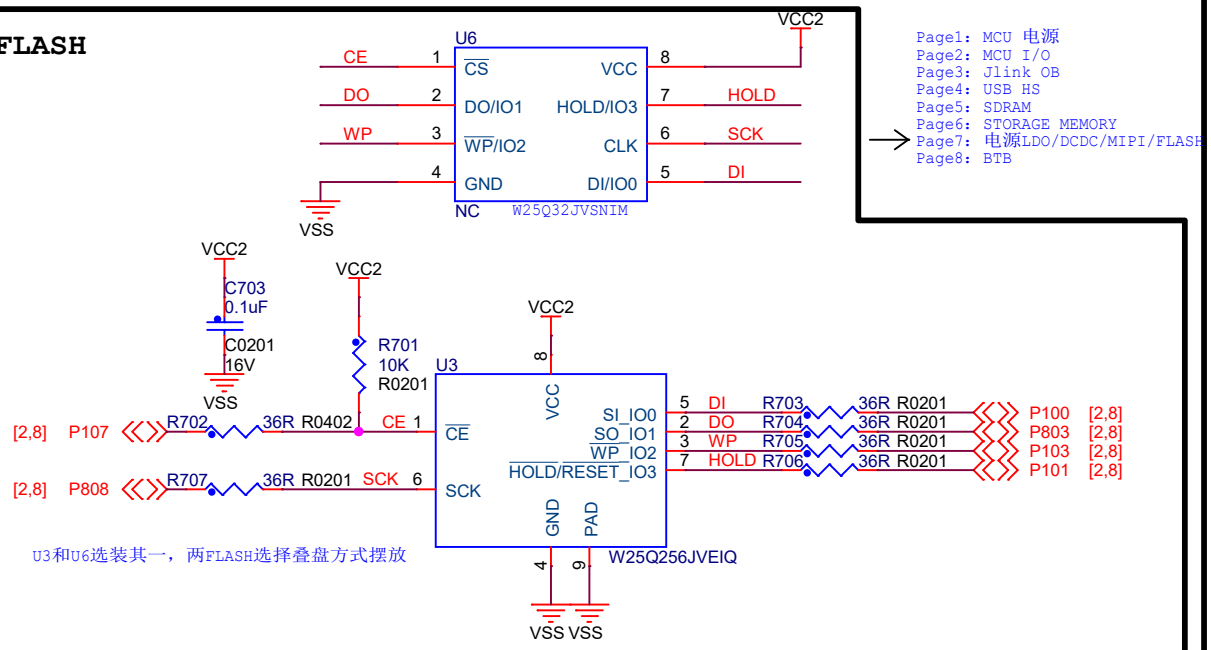


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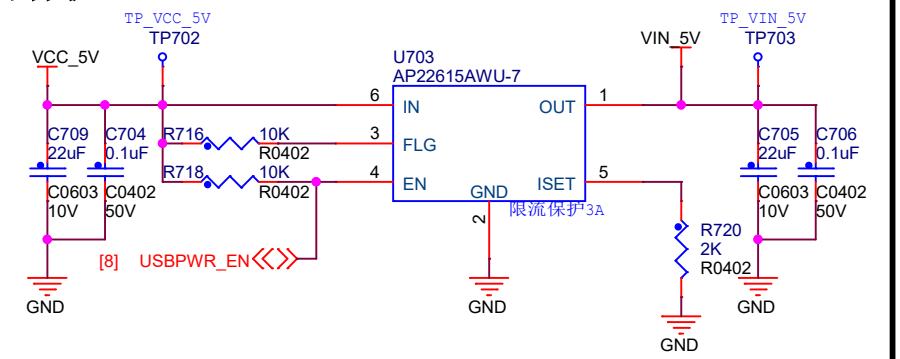
1.8V电源



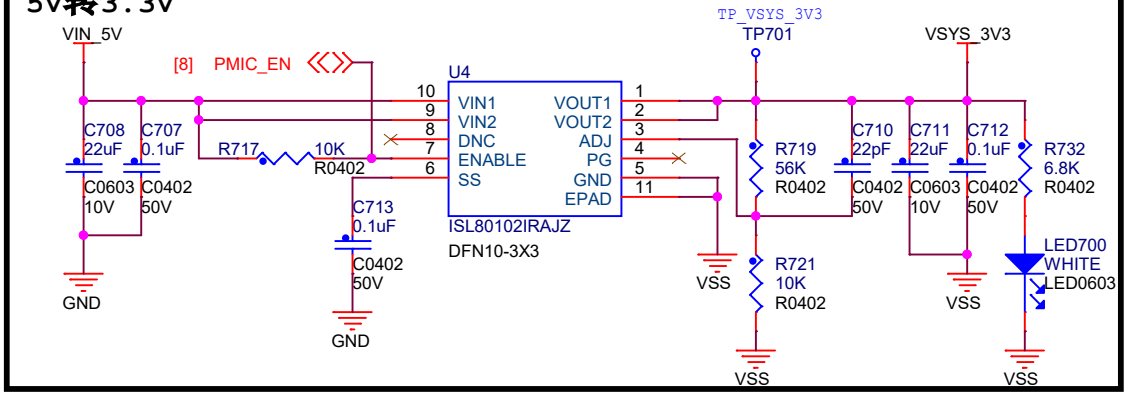
FLASH



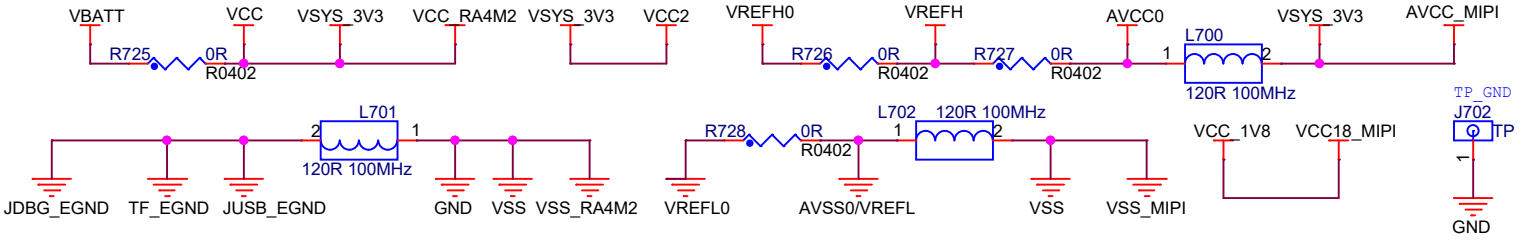
电源保护



5V转3.3V



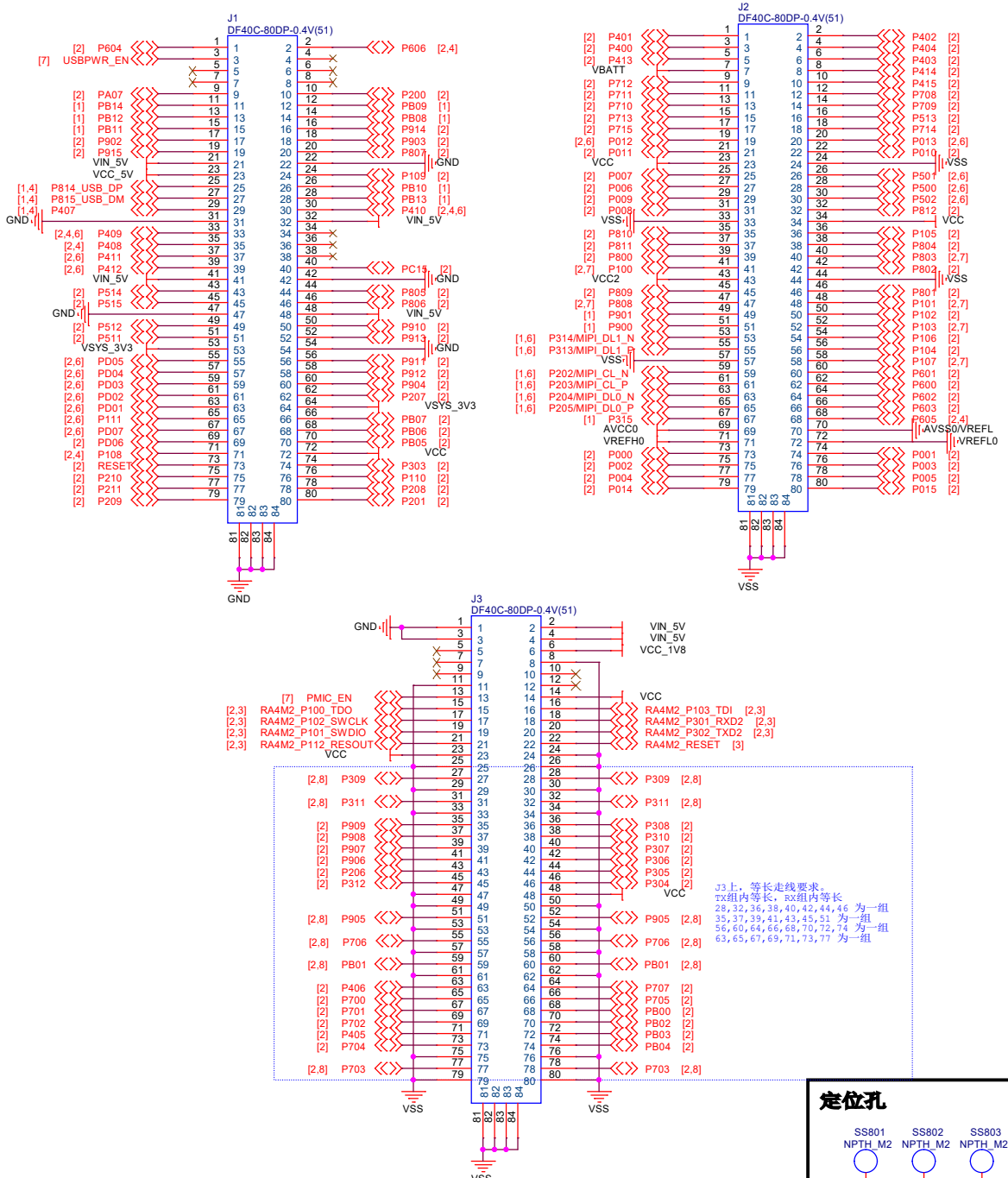
电源分割



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BTB座子

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历史版本

版本号	日期	设计	描述
V1.0	2025-08-28	LPC	量产版本
V1.1	2025-09-17	LPC	修正VCC_USB连接
V1.2	2026-06-08	llm	修改R205和R207两个电阻阻值

野火瑞萨CPKHMI-RA8P1K核心板_原理图_V1.2

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Title: 野火瑞萨CPKHMI-RA8P1K核心板

Size: A4 | Document Number: 目录 | Rev: V1.2

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